Locally Decodable Codes

Sayan Kar

Last updated

May 26, 2026

Abstract

Locally Decodable Codes as discussed in [Yek12].

NOTATIONS

$\mathbb{F}_q$	Finite field of size q
$\mathbb{F}_q^*$	Multiplicative group of $\mathbb{F}_q$
$\mathbb{Z}_m$	residue class of m
$\mathbb{N}$	Set of natural numbers
$[n]$	The set $\{1, 2, \ldots, n\}$
$\emptyset$	Empty set
$X \sim \mathscr{D}$	Random variable X having distribution $\mathscr{D}$
$\Delta(\cdot, \cdot)$	Relative Hamming distance
$\mathbb{E}[X]$	Expectation of random variable X
$\mathbb{D}[X]$	Variance of random variable X
$\mathbf{w}(l)$	l-th coordinate of $\mathbf{w} \in \mathbb{F}_q^n$
D-evaluation of h over D	Vector of values of h at all points of D
$\mathrm{Unif}(S)$	Uniform distribution over the set S
q	A prime power, unless stated otherwise
$g^{\mathbf{w}}$	$(g^{\mathbf{w}(1)}, \ldots, g^{\mathbf{w}(n)})$
$M_{\mathbf{w},\mathbf{v}}$	$\{g^{\mathbf{w}+\lambda\mathbf{v}} \mid \lambda \in \mathbb{Z}_m\}$
$\mathrm{mon}_{\mathbf{u}}(z_1, \ldots, z_n)$	$\prod_\ell z_\ell^{\mathbf{u}(\ell)}$ in $\mathbb{F}_q[z_1, \ldots, z_n]/(z_1^m - 1, \ldots, z_n^m - 1)$

$\langle \mathbf{u}, \mathbf{v} \rangle$	Inner product of vectors **u** and **v**
$\mathrm{supp}(h(y))$	$\{e \in \mathbb{N}_{\geq 0} : y^e$ has positive coefficient in $h\}$
b-bounded set	Every element of the set ($\subseteq \mathbb{R}$) is less than b
$\mathrm{agr}(e, M)$	Number of agreements of M-evaluation of F and C_m-evaluation of h
$\mathrm{weight}(e, M)$	$\max\{\mathrm{agr}(e, M) : h \in \mathbb{F}[y], h(0) = e, \mathrm{supp}(h) \subseteq S \cup \{0\}\}$
$A \otimes B$	Kronecker product of A and B
$A \odot B$	Hadamard Product of A and B
$\lVert \mathbf{a} \rVert$	Hamming weight of **a**

Contents

Chapter 1 INTRODUCTION

1.1 Definition of Local Decoding and Correction

This chapter mostly consists of *local correction* of some linear codes *along some curves*, mainly linear and quadratic. Let us first define what is local correction and decoding formally.

Definition 1.1 (Locally Decodable Code)**.** A q-ary code $C : \mathbb{F}_q^k \to \mathbb{F}_q^N$ is said to be (r, δ, ε)-*locally decodable* if there exists a randomized decoding algorithm $\mathscr{A}$ such that

1. For all $\mathbf{x} \in \mathbb{F}_q^k$, $i \in [k]$, and all vectors $\mathbf{y} \in \mathbb{F}_q^N$ satisfying $\Delta(C(\mathbf{x}), \mathbf{y}) \leq \delta$, we have:

$$\mathbb{P}\mathrm{r}[\mathscr{A}^{\mathbf{y}}(i) = \mathbf{x}(i)] \geq 1 - \varepsilon,$$

   where the probability is taken over the random coin tosses of the algorithm $\mathscr{A}$.

2. $\mathscr{A}$ makes at most r queries to $\mathbf{y}$.

Definition 1.2 (Locally Correctable Code)**.** A code $C \subseteq \mathbb{F}_q^N$ is said to be (r, δ, ε)-locally correctable if there exists a randomized correcting algorithm $\mathscr{A}$ such that

1. For all $\mathbf{c} \in C$, $i \in [N]$, and all vectors $\mathbf{y} \in \mathbb{F}_q^N$ satisfying $\Delta(\mathbf{c}, \mathbf{y}) \leq \delta$, we have

$$\mathbb{P}\mathrm{r}[\mathscr{A}^{\mathbf{y}}(i) = \mathbf{c}(i)] \geq 1 - \varepsilon,$$

   where the probability is taken over the random coin tosses of the algorithm $\mathscr{A}$.

2. $\mathscr{A}$ makes at most r queries to $\mathbf{y}$.

The main difference between them is that in locally decodable code, we want to retrieve coordinates of the sent message from recieved word and for locally correctable codes, we want to retrieve the coordinates of codewords. Lemma 2.3 in [Yek12] shows that a linear locally correctable code implies existence of a locally decodable code with the same parameter as of the first code.

1.2 Reed Muller locally decodable codes

Before going into the correction of Reed Muller, recall the definition of Reed Muller code.

Definition 1.3 (Reed Muller Codes)**.** For a finite field $\mathbb{F}_q$, define evaluation of $f \in \mathbb{F}[x_1, \ldots, x_n]$ at $\mathbf{z} = (z_1, \ldots, z_n) \in \mathbb{F}_q^n$ to be

$$\mathrm{Eval}_{\mathbf{z}}(f) = f(z_1, \ldots, z_n).$$

Also define $\mathrm{Eval}(f) := (Eval_{\mathbf{z}_1}(f), \ldots, Eval_{\mathbf{z}_{q^n}}(f)) \in \mathbb{F}_{q^n}$ for some fixed ordering in $\mathbb{F}_{q^n}$. Now define *Reed-Muller Code* of order r as

$$\mathrm{RM}(n, r) := \{\mathrm{Eval}(f) : f \in \mathbb{F}[x_1, \ldots, x_n], \deg f \leq r\}.$$

Our main goal of this section is to recover the evaluation of $f \in \mathbb{F}_q[z_1, \ldots, z_n]$ at $\mathbf{w} \in \mathbb{F}_q^n$ from evaluations, possibly corrupted, of $f \in \mathbb{F}_q[z_1, \ldots, z_n]$ at some points on a curve passing through $\mathbf{w}$. So for the most basic local correction, we take a line passing through $\mathbf{w}$ first, then move into degree 2 curve.

1.2.1 Basic Linear Local Correction

The main proposition governing this section is the following.

Proposition 1.4. *Let n and d be positive integers. Let q be a prime power, $d < q - 1$; then there exists a linear code of dimension $k = \binom{n+d}{d}$ in $\mathbb{F}_q^N$, $N = q^n$, that is $(d+1, \delta, (d+1)\delta)$-locally correctable for all δ.*

We shall find the failure probability, i.e., the ε among the three parameters. The corrector procedure is given fully in detail in Proposition 2.4 in [Yek12]. The main idea of the proof is to look at all $\mathbb{F}_q^n$ evaluations of a total degree d multivariate polynomial $f \in \mathbb{F}_q[z_1, \ldots, z_n]$. The goal of the linear corrector would be to retrieve $f(\mathbf{w})$ for a given $\mathbf{w} \in \mathbb{F}_q^n$, with only the recieved word $\tilde{\mathbf{c}}$ with high probability. Now what the corrector does is, it takes $d+1$ random points(uniform) on a random line passing through $\mathbf{w}$ and queries $\tilde{\mathbf{c}}$ at those evaluation points and find a univariate polynomial h of degree atmost d that is the restriction of f to that line. We shall show that the failure probability of the corrector is atmost $(d+1)\delta$.

Proof of failure probability. Consider $\mathbf{v} \sim \mathrm{Unif}(\mathbb{F}_q^n)$ and the line $L = \{\mathbf{w} + \lambda\mathbf{v} : \lambda \in \mathbb{F}_q\}$. Also consider a uniformly sampled subset S of $\mathbb{F}_q^*$ of size $d+1$. Now as the corrector queries the points, we observe that for all $\lambda \in S$,

$$\mathbb{P}\mathrm{r}[f(\mathbf{w} + \lambda\mathbf{v}) \neq c_{\mathbf{w}+\lambda\mathbf{v}}] \leq \delta.$$

Now the success probability is

$$\begin{aligned}
\mathbb{P}\mathrm{r}[f(\mathbf{w} + \lambda\mathbf{v}) = c_{\mathbf{w}+\lambda\mathbf{v}} \text{ for all } \lambda \in S] &= 1 - \mathbb{P}\mathrm{r}\left[f(\mathbf{w} + \lambda\mathbf{v}) \neq c_{\mathbf{w}+\lambda\mathbf{v}} \text{ for some } \lambda \in S\right] \\
&\geq 1 - \sum_{\lambda \in S} \mathbb{P}\mathrm{r}\left[f(\mathbf{w} + \lambda\mathbf{v}) \neq c_{\mathbf{w}+\lambda\mathbf{v}}\right] \\
&\geq 1 - (d+1)\delta.
\end{aligned}$$

□

1.2.2 Improvement on line local decoding

Now naturally we would want to decrease the failure probability, thus we would want to make the above decoding better. Here, keep the code same and try to interpolate the univariate restricition polynomial from fewer query. For this, we would need d to be very smaller that q. The main proposition for this section is the following:

Proposition 1.5. *Let $0 < \sigma < 1$, n and d be positive integers. Let q be a prime power such that $d \leq \sigma(q-1) - 1$; then there exists a linear code of dimension $k = \binom{n+d}{d}$ in $\mathbb{F}_q^N$, $N = q^n$, that is $(q-1, \delta, 2\delta/(1-\sigma))$-locally correctable for all δ.*

We use basically the same setup as before but change the interpolation part, here we query on atmost $(1-\sigma)(q-1)/2$ points and interpolate the polynomial via *Berlekamp-Welch* as given in [GRS25]. The reason for such a weird query number is because beyond this, Berlekamp-Welch can not uniquely interpolate the polynomial. As $d < \sigma(q-1)$, we have $(1-\sigma)(q-1)/2 > (q-1-d)/2$, this surpasses the RS bound for unique decoding.

Proof of failure probability. We shall calculate the failure a bit differently here, we shall do it via the probability of the random variable X denoting the number of corrupted query points is very big, i.e., $\mathbb{P}\mathrm{r}[\mathrm{X} \geq (1-\sigma)(q-1)/2]$. This we can get an upper bound by *Markov Inequality* as given in [AS00] if we can find $\mathbb{E}[\mathrm{X}]$. If we can find the expectation, the rest is just calculations that is done in the proof of Proposition 2.5 in [Yek12].

Notice that

$$\begin{aligned}
\mathbb{E}[\mathrm{X}] &= \mathbb{E}\left[\sum_{\lambda\in\mathbb{F}_q^*} \mathbb{1}_{\{f(\mathbf{w}+\lambda\mathbf{v}) \text{ is corrupted }\}}\right] \qquad (1.1)\\
&= \sum_{\lambda\in\mathbb{F}_q^*} \mathbb{P}\mathrm{r}[f(\mathbf{w}+\lambda\mathbf{v}) \text{ is corrupted }]\\
&\leq \sum_{\lambda\in\mathbb{F}_q^*} \delta = (q-1)\delta
\end{aligned}$$

This now gives the failure probability for the worst possible case:

$$\begin{aligned}
\mathbb{P}\mathrm{r}[\mathrm{X} \geq (1-\sigma)(q-1)/2] &\leq \frac{\mathbb{E}[\mathrm{X}]}{(1-\sigma)(q-1)/2}\\
&\leq \frac{2\delta}{1-\sigma}.
\end{aligned}$$

□

1.2.3 Decoding on Curves

Now we go from the line to a quadratic parametrized curve, i.e., we consider the curve $\chi = \{\mathbf{w} + \lambda\mathbf{v}_1 + \lambda^2\mathbf{v}_2 : \lambda \in \mathbb{F}_q^*\}$, where $\mathbf{v}_1, \mathbf{v}_2 \sim \mathrm{Unif}(\mathbb{F}_q^n)$.

Proposition 1.6. *Let $0 < \sigma < 1$, n and d be positive integers. Let q be a prime power such that $d \leq \sigma(q-1) - 1$; then there exists a linear code of dimension $k = \binom{n+d}{d}$ in $\mathbb{F}_q^N$, $N = q^n$, that for all positive $\delta < 1/2 - \sigma$ is $(q-1, \delta, O_{\sigma,\delta}(1/q))$-locally correctable.*

Here we shall take the same code and everything would be the same as Proposition 1.5 but the only difference would be that the interpolating polynomial is having degree atmost $2d$ and number of query points is atmost $(1-2\sigma)(q-1)/2$. We shall calculate the failure probability by the same random variable here also but in a different way, and instead of using the Markov inequality, we shall be using Chebyshev Inequality, given in [AS00].

Proof of failure probability. We want to find the failure probability, that is

$$\mathbb{P}\mathrm{r}\left[X \geq \frac{1}{2}(1-2\sigma)(q-1)\right].$$

Notice that the indicator random variables in 1.1 are independent, thus the variance of X becomes $\mathbb{D}[X] \leq (\delta - \delta^2)(q-1)$. So by applying Chebyshev, we get failure probability in worst case scenario.

$$\begin{aligned}\mathbb{P}\mathrm{r}\left[X \geq \frac{1}{2}(1-2\sigma)(q-1)\right] &\leq \frac{\mathbb{D}[X]}{((1-2\sigma)(q-1)/2 - (q-1)\delta)^2} \\ &= O_{\delta,\sigma}(1/q).\end{aligned}$$

□

1.3 Binary Code Correction

In this section, we shall use whatever idea we have developed so far and use Code Concetanation, given in [GRS25], to locally correct binary codes. Recall the definition of code concetanation.

Definition 1.7. Code Concetanation For $q \geq 2$, $k \geq 1$ and $Q = q^k$, consider two codes which we call outer code and inner code:

$$C_{\text{out}} : [Q]^K \to [Q]^N,$$
$$C_{\text{in}} : [q]^k \to [q]^n.$$

Note that the alphabet size of C_{out} exactly matches the number of messages for C_{in}, which means that we can have a bijection between $[Q]$ and $[q]^k$ (this means we can use C_{in} to encode any symbol in a codeword in C_{out}). Then given $m = (m_1, \ldots, m_K) \in [Q]^K$, we have the code $C_{\text{out}} \circ C_{\text{in}} : [q]^{kK} \to [q]^{nN}$ defined as

$$C_{\text{out}} \circ C_{\text{in}}(m) = \left(C_{\text{in}}(C_{\text{out}}(m)_1), \ldots, C_{\text{in}}(C_{\text{out}}(m)_N)\right),$$

where

$$C_{\text{out}}(m) = \left(C_{\text{out}}(m)_1, \ldots, C_{\text{out}}(m)_N\right).$$

The main proposition of this section is:

> **Proposition 1.8.** *Let $0 < \sigma < 1$, n and d be positive integers. Let $q = 2^b$ be a power of two such that $d \leq \sigma(q-1) - 1$. Suppose that there exists a binary linear code C_{in} of distance μB encoding b-bit messages to B-bit codewords; then there exists a linear code C of dimension $k = \binom{n+d}{d} \cdot b$ in $\mathbb{F}_2^N$, $N = q^n \cdot B$, that for all positive $\delta < (1/2 - \sigma)\mu$ is $((q-1)B, \delta, O_{\sigma,\mu,\delta}(1/q))$-locally correctable.*

Here we take the code to be the code concetanation of C_{in} and the code we discussed in the previous section, with C_{in} as the inner code. Then we want to find the inetrpolating polynomial h of degree atmost $2d$ by seeing agreement of h and C_{in} encoding of h by atmost $(1-2\sigma)(q-1)\mu B/2$ queries.

Proof of failure probability. We want to decompose the same random variable X by weighted sum of the same indicator random variables as in 1.1. For $\mathbf{a} \in \mathbb{F}_q^n$, define $t_{\mathbf{a}}$ to be the number of corruptions in the C_{in} encoding of $\mathrm{Eval}_{\mathbf{a}}(f)$, so we have $\sum_{\mathbb{F}_q^n} t_{\mathbf{a}} \leq \delta B q^n$. Now for $\lambda \in \mathbb{F}_q^*$,

$$X^\lambda = \sum_{\mathbf{a} \in \mathbb{F}_q^n} t_{\mathbf{a}} \cdot \mathbb{1}_{\{\chi(\lambda) = \mathbf{a}\}}.$$

So total number of corrupted query points is $X = \sum_{\lambda \in \mathbb{F}_q^*} X^\lambda$. Now the failure probability that we want to calculate is

$$\mathbb{P}\mathrm{r}\left[X \geq \frac{1}{2}(1 - 2\sigma)(q-1)\mu B\right].$$

Now similary we get that $\mathbb{E}[X] \leq \delta(q-1)B$ and $\mathbb{D}[X] \leq (\delta - \delta^2)(q-1)B^2$, hence applying Chebyshev, we get out desired bound for the failure probability. $\square$

Chapter 2 MATCHING VECTOR CODES

2.1 Introduction

In the RM codes seen in Chapter 1, we could do local correction upto $\delta < 1/4$ fraction of error tolerance. Our main goal of this chapter would be to construct a code that can tolerate upto $\delta < 1/2$ fraction of errors, which would be optimal for unique local decoding.

We first define what *matching vectors* are, then we can specify the encoding of *matching vector code*.

Definition 2.1 (Matching Vectors)**.** Let $S \subseteq \mathbb{Z}_m^n$, we say that the families $\mathcal{U} = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $\mathcal{V} = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of vectors in $\mathbb{Z}_m^n$ forms an *S-matching family* if

1. For all $i \in [k]$, $(\mathbf{u}_i, \mathbf{v}_i) = 0$,
2. For all $i \neq j$ in $[k]$, $(\mathbf{u}_j, \mathbf{u}_i) \in S$.

Observation 2.2. Let g be a generator of the unique subgroup C_m of $\mathbb{F}_q^*$ of size m. Then for any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{Z}_m^n$ and $\lambda \in \mathbb{Z}_m$,

$$\begin{aligned}
\mathrm{mon}_{\mathbf{u}}(g^{w+\lambda v}) &= \prod_{l=1}^{n} \left(g^{\mathbf{w}(l)+\lambda \mathbf{v}(l)}\right)^{\mathbf{u}(l)} \\
&= \prod_{l=1}^{n} g^{\mathbf{w}(l)\mathbf{u}(l)} \cdot \prod_{l=1}^{n} (g^{\lambda})^{\mathbf{v}(l)\mathbf{u}(l)} \\
&= g^{(\mathbf{u},\mathbf{w})} \cdot (g^{\lambda})^{(\mathbf{v},\mathbf{u})}.
\end{aligned}$$

So we can see that $M_{\mathbf{w},\mathbf{v}}$-evaluation of $\mathrm{mon}_{\mathbf{u}}$ is a C_m-evaluation of a univariate polynomial $g^{(\mathbf{u},\mathbf{w})} \cdot y^{(\mathbf{v},\mathbf{u})} \in \mathbb{F}_q[y]$.

Now we define the encoding of matching vector codes, but we first need to assume that there exists an S-matching family $\mathcal{U}$ and $\mathcal{V}$ of size k. We shall deal with this existence later. There is also an abstract decoding method mentioned that will be the main motivation behind all the decoding procedure in the coming sections.

(ENCODING) We take a k-length message $\mathbf{x} = (\mathbf{x}(1), \ldots, \mathbf{x}(k)) \in \mathbb{F}_q^k$ and encode ot by C_m^n-evaluations of the polynomial

$$F_{\mathbf{x}}(z_1, \ldots, z_n) = \sum_{j=1}^{k} x(j)\mathrm{mon}_{u_j}(z_1, \ldots, z_n). \tag{2.1}$$

(ABSTRACT DECODING) We want to retrieve $\mathbf{x}(i)$ for a specified index $i \in [k]$. The decoding procedure is:

1. Pick $\mathbf{w} \sim \mathrm{Unif}(\mathbb{Z}_m^n)$.
2. Compute the restriction of $F \equiv F_{\mathbf{x}}$ to $M_{\mathbf{w},v_i}$ by some queries of the corrupted M_{w,v_i}-evaluation of F at less than or equal to m points.

One might ask now that what is the importance of this restriction and how it will help in retrieving $\mathbf{x}(i)$. The following observation gives us the motivation of using matching family and this second step.

Observation 2.3. The M_{w,v_i}-evaluation of F is equivalent to the C_m^n-evaluation of the polynomial f defined as:

$$f(y) = \sum_{j=1}^{k} x(j) g^{(\mathbf{u}_j, \mathbf{w})} \cdot y^{(\mathbf{v}_i, \mathbf{u}_j)}$$

By matching family property, we can write the polynomial as:

$$f(y) = x(i) g^{(\mathbf{u}_i, \mathbf{v}_i)} + \sum_{s \in S} \left(\sum_{j:(\mathbf{u}_j, \mathbf{v}_i) = s} x(j) g^{(\mathbf{u}_j, \mathbf{v}_i)} \right) y^s.$$

So we can get $\mathbf{x}(i)$ by just computing $f(0)/g^{(\mathbf{u}_i, \mathbf{w})}$. But of course we can not construct this polynomial, so we try to find the restriction of F to $M_{\mathbf{w},\mathbf{v}_i}$.

2.2 Some Decoding Procedures

We shall now see some explicit decoding procedures along with making the error tolerance better and better.

2.2.1 Basic Decoding on Lines

Proposition 2.4. *Let $\mathcal{U}, \mathcal{V}$ be a family of S-matching vectors in $\mathbb{Z}_m^n$, with $|\mathcal{U}| = |\mathcal{V}| = k$ and $|S| = s$. Suppose $m \mid (q-1)$, where q is a prime power. Then there exists a q-ary linear code encoding k-long messages to m^n-long codewords that is $(s+1, \delta, (s+1)\delta)$-locally decodable for all δ.*

Consider the code specified in Equation 2.1. Now do the following decoding procedure:

1. Pick $\mathbf{w} \sim \mathrm{Unif}(\mathbb{Z}_m^n)$ and construct the multiplicative line $M_{\mathbf{w},\mathbf{v}_i}$.
2. Query the M-evaluation of F at $s+1$ many consecutive locations, say $\{g^{\mathbf{w}+\lambda\mathbf{v}_i} : \lambda \in \{0, \ldots, s\}\}$ and obtain the points $c_0, \ldots, c_s$.
3. Interpolate the unique sparse polynomial $h \in \mathbb{F}_q[y]$ with $\mathrm{supp}(h) \subseteq S \cup \{0\}$ such that $h(g^\lambda) = c_\lambda$ for all λ.
4. Output $h(0)/g^{(\mathbf{w},\mathbf{u}_i)}$.

Proof for failure probability. This is a fairly easy probability to calculate.

$$\begin{aligned}\mathbb{P}\mathrm{r}[\text{failure of decoder}] &= \mathbb{P}\mathrm{r}[F(g^{\mathbf{w}+\lambda\mathbf{v}_i}) \neq c_\lambda \text{ for some } \lambda] \\ &\leq \sum_\lambda \mathbb{P}\mathrm{r}[F(g^{\mathbf{w}+\lambda\mathbf{v}_i}) \neq c_\lambda] \\ &\leq \sum_\lambda \delta = (s+1)\delta.\end{aligned}$$

□

2.2.2 Improvement on Line Decoding

In this section we shall see how we can make the line decoding better to get error tolerance of $\delta < 1/4$ by making the matching families *bounded*.

Definition 2.5 (b-bounded Matching Vectors)**.** $\mathcal{U}, \mathcal{V}$ are said to be a b*-bounded* S-matching family over $\mathbb{Z}_m^n$ if S is b-bounded.

We now give the main proposition of this subsection analogous to Proposition 1.5.

Proposition 2.6. *Let $0 < \sigma < 1$. Let $\mathcal{U}, \mathcal{V}$ be a σm-bounded family of S-matching vectors in $\mathbb{Z}_m^n$, $|\mathcal{U}| = |\mathcal{V}| = k$. Suppose $m \mid (q-1)$, where q is a prime power. Then there exists a q-ary linear code encoding k-long messages to m^n-long codewords that is $(m, \delta, 2\delta/(1-\sigma))$-locally decodable for all δ.*

Consider the same code as before and the following decoding procedure:

1. Pick $\mathbf{w} \sim \mathrm{Unif}(\mathbb{Z}_m^n)$ and construct the multiplicative line $M_{\mathbf{w},\mathbf{v}_i}$.
2. Query the M-evaluation of F at all of the m locations, and obtain the points $c_0, \ldots, c_{m-1}$.
3. By Berlekamp-Welch algorithm, given in [GRS25], compute the unique univariate polynomial $h(y) \in \mathbb{F}_q[y]$ of degree less that σm and $\mathrm{supp}(h) \subseteq S \cup \{0\}$ such that h agrees with the corrupted M-evaluation of F at all but at most $m(1-\sigma)/2$ many points.
4. Return $h(0)/g^{(\mathbf{u}_i,\mathbf{w})}$ if such h exists, else return 0.

Proof for failure probability. Take the random variable X denoting the number of corrupted query points in the M-evaluation of F. Now failure occurs if $X \geq m(1-\sigma)/2$, hence failure probability is less than or equal to:

$$\begin{aligned}
\Pr\left[X \geq \frac{m(1-\sigma)}{2}\right] &\leq \frac{1}{m(1-\sigma)/2}\mathbb{E}[X] \quad \text{[Markov Inequality]} \\
&= \frac{1}{m(1-\sigma)/2}\sum_{\lambda \in \mathbb{Z}_m} \Pr\left[F(g^{\mathbf{w}+\lambda\mathbf{v}_i}) \neq c_\lambda\right] \\
&\leq \frac{2m\delta}{m(1-\sigma)} = \frac{2\delta}{1-\sigma}.
\end{aligned}$$

$\square$

Observation 2.7 (Error Tolerance)**.** As $\sigma \downarrow 0$, the limiting failure probability is 2δ, hence we can only tolerate up to $\delta < 1/4$ fraction of errors, because for error tolerance, the failure probability must be less that $1/2$. In the next subsection, we show a decoding paradigm that can tolerate up to $1/2$ fraction of errors.

2.2.3 Decoding on Multiple Lines

Rather than constructing one random multiplicative line, we shall construct more and using the queries at each M-evaluation of F, we shall interpolate the polynomial.

> **Proposition 2.8.** *Let $0 < \sigma < 1$. Let $\mathcal{U}, \mathcal{V}$ be a σm-bounded family of S-matching vectors in $\mathbb{Z}_m^n$, $|\mathcal{U}| = |\mathcal{V}| = k$. Suppose $m \mid (q-1)$, where q is a prime power. Then for every positive integer l there exists a q-ary linear code encoding k-long messages to m^n-long codewords that for all $\delta < (1-\sigma)/2$ is $(lm, \delta, \exp_{\delta,\sigma}(-l))$-locally decodable.*

Consider the same code as before and fix an $l \in \mathbb{N}$. Now here we shall not construct the polynomial like we have do before, we shall consider l many M-evaluations of F and instead of interpolating the polynomial directly, we shall assign weight to each field element e with respect to the likelihood of $x(i) = e$. And from that we shall return the unique highest weight field element, if exists. The decoding procedure is the following:

1. Take $\mathbf{w}_1, \ldots, \mathbf{w}_l$ uniformly at random from $\mathbb{Z}_m^n$ and consider the l many multiplicative lines $\{M_{\mathbf{w}_j, \mathbf{v}_i} : j \in [l]\}$.

2. Query at the lm many coordinates of all the M-evaluations of F and obtain l many vectors in $\mathbb{F}_q^m$, say $\mathbf{c}^1, \ldots, \mathbf{c}^m$. For each $e \in \mathbb{F}_q$, compute:

$$\text{weight}(e) = \sum_{j=1}^{l} \text{weight}(e, M_{\mathbf{w}_j, \mathbf{v}_i}).$$

3. Return e with the highest weight. If it is not unique, then return 0.

Observation 2.9. Suppose there are two distinct $e_1, e_2 \in \mathbb{F}_q$ such that $\text{weight}(e_1), \text{weight}(e_2) >$

$lm(1+\sigma)/2$. Then by PHP, there would be two distinct polynomials $h_1, h_2 \in \mathbb{F}_q[y]$, both of degree less than σm and two multiplicative lines M_1, M_2 such that weight$(e_1, M_1) > m(1+\sigma)/2$ and weight$(e_2, M_2) > m(1+\sigma)/2$. Now by inclusion-exclusion, the numeber of agreement points of h_1 and h_2 is at least $2m(1+\sigma)/2 - m = \sigma m$. Thus we can not have such a pair.

Proof for failure probability. Consider the random variable X denoting the number of corrupted queried points and failure probability is less than or equal to $\mathbb{P}\mathrm{r}[X \geq lm(1+\sigma)/2]$. Note that,

$$X = \sum_{j=1}^{l} \left(\sum_{\lambda \in \mathbb{Z}_m} \mathbb{1}_{\{F(g_\lambda^{(\mathbf{w}_j, \mathbf{v}_i)}) \neq \mathbf{c}_\lambda^j\}} \right)$$

Now the expected value of X is:

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{j=1}^{l} \left(\sum_{\lambda \in \mathbb{Z}_m} \mathbb{P}\mathrm{r}[F(g_\lambda^{(\mathbf{w}_j, \mathbf{v}_i)}) \neq \mathbf{c}_\lambda^j] \right) \\
&\leq \sum_{j=1}^{l} \left(\sum_{\lambda \in \mathbb{Z}_m} \delta \right) = ml\delta.
\end{aligned}$$

Now X is the sum of lm many independent and identically distributed Bernoulli random variable of success probability δ. Now we can apply *Chernoff bound*, given in [AS00], in the following way to find an upper bound for failure probability. Using $\delta < (1-\sigma)/2$,

$$\begin{aligned}
\mathbb{P}\mathrm{r}[X \geq lm(1+\sigma)/2] &\leq \mathbb{P}\mathrm{r}[X - \mathbb{E}[X] \geq lm(1+\sigma)/2 - lm(1-\sigma)/2] \\
&= \mathbb{P}\mathrm{r}[X - \mathbb{E}[X] \geq ml\sigma] \\
&\leq \exp_{\sigma,\delta}(-l) \quad \text{Chernoff Bound.}
\end{aligned}$$

□

Now we can show how we increase the error tolerance up to $\delta < 1/2$. Since the proposition holds for every $\delta < (1-\sigma)/2$, letting $\sigma \downarrow 0$ yields $\delta \uparrow 1/2$. Hence the decoder tolerates any error fraction $\delta < 1/2$.

Chapter 3 MATCHING VECTORS

Recall the definition of *matching vectors*, given in Definition 2.1. We shall give some combinatorial and algorithmic viewpoint for constructing such matching vector families in this chapter. The main references for this are [MBBR92] and [Gro00].

3.1 Grolmusz Theorem

This section shall mainly consist of the construction of an MVF as per [Gro00]. The main result of Grolmusz is the following:

Theorem 3.1. *Let $m = \prod_{i=1}^{t} p_i$ be a product of distinct primes. Then there exists an explicit matching vector family in $\mathbb{Z}_m^l$ of size*

$$k = \exp\left(\Omega\left(\frac{(\log l)^t}{(\log\log l)^{t-1}}\right)\right).$$

Before going to the actual proof of this theorem, we first state some interesting ideas. One of the most important combinatorial question that can arise in one's mind is how large the size of the family can be. The following lemma gives an upper bound on the size of an MVF.

Lemma 3.2. *If p is a prime, then any MVF in $\mathbb{Z}_m^l$ has size at most $1 + l^{p-1}$.*

Proof. Let $\mathcal{U}, \mathcal{V}$ be a MVF in $\mathbb{Z}_p^l$ of size k. Consider the matrix $A \in \mathbb{Z}_p^{k\times k}$ with $A_{i,j} = \langle \mathbf{u}_i, \mathbf{v}_j \rangle$. By *Fermat's Little Theorem*, Lemma A.3 in Appendix A, we observe that $A_{i,j}^{p-1} = 1$ for all i, j. Observe, $A^{p-1} = J_k - I_k$, hence $\mathsf{rank}(A^{p-1}) \leq k - 1$ and $\mathsf{rank}(A^{p-1}) \leq \mathsf{rank}(A)^{p-1} \leq l^{p-1}$, by Lemma A.2. $\square$

3.1.1 Polynomial Representation of OR-function in mod 6

Definition 3.3 (OR Function)**.** n-variate OR function, denoted as OR_n, is a boolean function defined as:

$$\mathrm{OR}_n(x_1, \ldots, x_n) = 0 \text{ if and only if } x_i = 0 \text{ for all } i \in [n].$$

We shall now study some polynomial "representative"(s) of this OR function.

Definition 3.4. A polynomial $g \in \mathbb{Z}_m[x_1, \ldots, x_n]$ *represents* a boolean function $f : \{0,1\}^n \to \{0,1\}$ mod m if for all $\mathbf{x} \in \{0,1\}^n$,

$$g(\mathbf{x}) \begin{cases} = 0 \pmod{m} & \text{if } f(\mathbf{x}) = 0 \\ \neq 0 \pmod{m} & \text{if } f(\mathbf{x}) = 1 \end{cases}$$

And the *degree* of the boolean function f modulo p, denoted as $\deg_p(f)$, is defined as $\min\{\deg(g) : g \text{ represents } f \text{ modulo } p\}$.

The following lemma shall give the main motivation behind the study of representative polynomials modulo a composite number with at least two distinct prime factors, instead of a prime power, or equivalently, shifting our focus from fields to rings!

Lemma 3.5. *If p is a prime, then* $\deg_p(\mathrm{OR}_n) \geq n/(p-1)$.

Proof. Let g be a polynomial representative of OR_n in $\mathbb{F}_p$, i.e., $\mathrm{OR}_n(\mathbf{a}) = 0$ if and only if $g(\mathbf{a}) = 0 \pmod{p}$. Notice that as we are working $\mathbb{F}_p$, we can transform g to make it behave like a boolean function in $\mathbb{F}_p$. Consider the polynomial $h(\mathbf{x}) = 1 - g^{p-1}(\mathbf{x})$. Now notice that as we are in $\{0,1\}^n$, we can make this h multilinear, say $\tilde{h}$. Observe, by Lemma A.3, $\tilde{h}(\mathbf{a}) = 1$ at $\mathbf{0}$ and 0 everywhere else. Also by *Lemma A.6*, we get that:

$$2^n - 1 = |\{\mathbf{a} \in \{0,1\}^n : \tilde{h}(\mathbf{a}) = 0\}| \leq 2^n - 2^{n - \deg \tilde{h}}.$$

Hence, $\deg \tilde{h} \geq n$ and with that we get our final bound:

$$n \leq \deg \tilde{h} \leq \deg h \leq (p-1) \deg g.$$

□

Also note we can extend Lemma 3.5 to prime powers, the only difference that would rise it taken care by Lemma A.3. Hence, we see that taking prime power ring, or the field, we get the minimum degree to be $\Omega(n)$ but [MBBR92] showed that for *product of distinct primes* size ring, we can get much smaller degree representative polynomials. We shall only see the proof of BBR theorem for modulo 6 but this can be extended to any m that is a product of distinct primes with the same principles used in the following proof.

3.1.2 Barrington-Beigel-Rudich Theorem

In this subsection, we shall show that using $\mathbb{Z}_6$ as the base ring for the polynomials, we can reduce the degree of OR_n function to $O(\sqrt{n})$.

> **Theorem 3.6.** *[Barrington-Beigel-Rudich] There exists an n-variate polynomial that represents* OR_n *modulo* 6 *with degree at most* $O(\sqrt{n})$*.*

Before going to this involved proof, let me give an intuitive proof outline of this theorem. Basically, we want to construct a polynomial $f \in \mathbb{Z}_6[x_1, \ldots, z_n]$ of degree at most $O(\sqrt{n})$ such that $\mathrm{OR}_n(\mathbf{a}) = 0$ if and only if $f(\mathbf{a}) = 0 \pmod 6$. Suppose we can construct two polynomials $f_2 \in \mathbb{Z}_2[x_1, \ldots, z_n]$ and $f_3 \in \mathbb{Z}_3[x_1, \ldots, z_n]$ such that for some $2^d \approx \sqrt{n}$ and $3^e \approx \sqrt{n}$, we have:

$$f_2(a) = 0 \iff \|a\| \equiv 0 \pmod{2^d} \quad \text{and} \quad f_3(a) = 0 \iff \|a\| \equiv 0 \pmod{3^e},$$

making sure that $\deg f_i$ is nearly $\sqrt{n}$. Intuitively this is possible due to the fact that the number of $\mathbf{a}$ that satisfies the above condition is at most $\sum_{i=1}^{\sqrt{n}} \binom{n}{i\sqrt{n}}$, which is very smaller than $\sqrt{n}2^{n-1}$, which is the *Schwarz-Zippel* bound, see Lemma A.5. Having this done, let us consider the isomorphism given in Theorem A.8. There exists a unique polynomial $f \in \mathbb{Z}_6[x_1, \ldots, x_n]$ such that $f_2 = f \pmod 2$ and $f_3 = f \pmod 3$. If we make sure that $2^d 3^e > n$, then we can show that f represents OR_n by the following arguements and Theorem A.7.

$$f(\mathbf{a}) = 0 \iff f_2(\mathbf{a}) = 0 \text{ and } f_3(\mathbf{a}) = 0 \iff \|\mathbf{a}\| = 0 \pmod{2^d 3^e} \iff \mathbf{a} = \mathbf{0},$$

because $\|\mathbf{a}\| \leq n$. And to see the degree of f, note that $\deg f \leq \max\{\deg f_2, \deg f_3\} = O(\sqrt{n})$.

Lemma 3.7. *If p is fixed prime, then for all $d \in \mathbb{N}$, there exists a polynomial $f_p^d \in \mathbb{Z}_p[x_1, \ldots, x_n]$ such that $\deg(f_p^d) \leq p^d - 1$ and $f_p^d(\mathbf{a}) = 0$ if and only if $\|\mathbf{a}\| = 0 \pmod{p^d}$.*

Proof. shall add later ... :) $\square$

Now we can prove Theorem 3.6 formally.

Proof of Theorem 3.6. Take d and e such that $\sqrt{n} < 2^d \leq 2\sqrt{n}$ and $\sqrt{n} < 3^e \leq 3\sqrt{n}$. By Lemma 3.7, there are two polynomials $f_2 \in \mathbb{Z}_2[x_1, \ldots, x_n]$ and $f_3 \in \mathbb{Z}_3[x_1, \ldots, x_n]$ such that $\deg(f_2) \leq 2^d - 1$, $\deg(f_3) \leq 3^e - 1$, and

$$f_2(a) = 0 \iff \|a\| \equiv 0 \pmod{2^d}, \qquad f_3(a) = 0 \iff \|a\| \equiv 0 \pmod{3^e}.$$

Now by Theorem A.8, $f = (f_2, f_3) \in \mathbb{Z}_6[x_1, \ldots, x_n]$ has degree at most $\max(\deg(f_2), \deg(f_3))$, which is at most $3\sqrt{n}$, and $f(a) = 0$ if and only if $\|a\| \equiv 0 \pmod{2^d 3^e}$. Since $2^d 3^e > n$, we have $\|a\| \equiv 0 \pmod{2^d 3^e}$ if and only if $\|a\| = 0$ if and only if for all $i \in [n]$, $a_i = 0$. So $f(a) = 0$ if and only if for all $i \in [n]$, $a_i = 0$. $\square$

3.2 Construction of MVF

Bibliography

[AS00] Noga Alon and Joel H. Spencer. *The Probabilistic Method*. Wiley, August 2000.

[Bur11] David M. Burton. *Elementary Number Theory*. McGraw-Hill, 7 edition, 2011.

[Gro00] Vince Grolmusz. Superpolynomial size set-systems with restricted intersections mod 6 and explicit ramsey graphs. *Combinatorica*, 20(1):71–86, January 2000.

[GRS25] Venkatesan Guruswami, Atri Rudra, and Madhu Sudan. *Essential Coding Theory*. August 2025. Draft manuscript, version of August 26, 2025. Available at https://cse.buffalo.edu/faculty/atri/courses/coding-theory/book/.

[MBBR92] David A. Mix Barrington, Richard Beigel, and Steven Rudich. Representing boolean functions as polynomials modulo composite numbers. In *Proceedings of the Twenty-Fourth Annual ACM Symposium on Theory of Computing*, STOC '92, page 455–461, New York, NY, USA, 1992. Association for Computing Machinery.

[Yek12] Sergey Yekhanin. Locally decodable codes. *Found. Trends Theor. Comput. Sci.*, 6(3):139–255, October 2012.

Chapter A APPENDIX

Lemma A.1. *If X, Y are two matrices over $\mathbb{F}_q$ of proper dimension, then* $\mathsf{rank}(X \otimes Y) = \mathsf{rank}(X) \cdot \mathsf{rank}(Y)$.

Lemma A.2. *Suppose $k, t \in \mathbb{N}$ and $A, B \in \mathbb{F}_q^{k \times k}$ such that $B_{i,j} = A_{i,j}^t$ for all $i, j \in [k]$. Then* $\mathsf{rank}(B) \leq \mathsf{rank}(A)^t$.

Proof. Take the matrix $T = \otimes^t A$. Now notice that B is a sub-matrix of T. By Lemma A.1, $\mathsf{rank}(B) \leq \mathsf{rank}(T) = \mathsf{rank}(A)^t$. □

Lemma A.3 (Fermat's Little Theorem [Bur11])**.** *If $a \in \mathbb{Z}$ and $p \in \mathbb{P}$ such that $\gcd(a, p) = 1$, then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Theorem A.4 (Lucas' Theorem [Bur11])**.** *Let p be a prime and let m, n be non-negative integers with base-p representations $m = \sum_i m_i p^i$ and $n = \sum_i n_i p^i$. Then*

$$\binom{m}{n} \equiv \prod_i \binom{m_i}{n_i} \pmod{p}$$

Lemma A.5 (Schwarz–Zippel)**.** *Let $f \in \mathbb{F}[x_1, \ldots, x_n]$ be a non-zero polynomial of total degree d over a field $\mathbb{F}$, and let $S \subseteq \mathbb{F}$ be a finite set. If $r_1, \ldots, r_n$ are chosen independently and uniformly at random from S, then:*

$$\Pr[f(r_1, \ldots, r_n) = 0] \leq \frac{d}{|S|}.$$

Lemma A.6. *Suppose $f \in \mathbb{F}_p[x_1, \ldots, x_n]$, where p is prime, is a multilinear non-zero polynomial of* *degree at most d, where $d > 0$. Then:*

$$|\{a \in \{0,1\}^n \mid f(a) = 0\}| \leq 2^n - 2^{n-d},$$

where equality is achieved for $f = \prod_{i=1}^{d} x_i$.

Proof. We can prove the lemma by induction on n. In the base case, $n = 1$, we have $d = 1$, so f is an expression linear in x_1, and thus can have at most one zero, so the inequality holds. Suppose we have shown the inequality holds for polynomials of $n-1$ variables. We show that it also holds for any polynomial f of n variables. We can express f in the following way:

$$f(x_1, \ldots, x_n) = x_1 g(x_2, \ldots, x_n) + h(x_2, \ldots, x_n).$$

Notice that the polynomial g is of degree at most $d-1$. Then, we have:

$$\begin{aligned}
|\{a \in \{0,1\}^n \mid f(a) = 0\}| &\leq 2|\{a \in \{0,1\}^{n-1} \mid g(a) = 0\}| + |\{a \in \{0,1\}^{n-1} \mid g(a) \neq 0\}| \\
&= |\{a \in \{0,1\}^{n-1} \mid g(a) = 0\}| + 2^{n-1} \\
&\leq 2^{n-1} - 2^{n-1-(d-1)} + 2^{n-1} \\
&= 2^n - 2^{n-d},
\end{aligned}$$

where the first inequality holds because if $g(x_2, \ldots, x_n) \neq 0$, there is exactly one value of x_1 for which $f(x) = 0$, but if $g(x_2, \ldots, x_n) = 0$, there might be up to two values of x_1 such that $f(x) = 0$. The second inequality holds by the inductive step. □

Theorem A.7 (Chinese Remainder Theorem for Integer Ring)**.** *Let p and q be prime numbers with $m = pq$. Then:*

$$\mathbb{Z}_m \cong \mathbb{Z}_p \times \mathbb{Z}_q,$$

where the isomorphism $\varphi : \mathbb{Z}_m \to \mathbb{Z}_p \times \mathbb{Z}_q$ is given by

$$\varphi(a) = (a \bmod p,\ a \bmod q).$$

Theorem A.8 (Chinese Remainder Theorem for Polynomial Rings)**.** *Let p and q be prime numbers with $m = pq$. Then:*

$$\mathbb{Z}_m[x_1, \ldots, x_n] \cong \mathbb{Z}_p[x_1, \ldots, x_n] \times \mathbb{Z}_q[x_1, \ldots, x_n],$$

where the isomorphism $\varphi : \mathbb{Z}_m[x_1, \ldots, x_n] \to \mathbb{Z}_p[x_1, \ldots, x_n] \times \mathbb{Z}_q[x_1, \ldots, x_n]$ is given by

$$\varphi(f(x)) = (f(x) \bmod p,\ f(x) \bmod q).$$

This works because a polynomial in n variables modulo some number t is an expression built from sums and products, so φ can be applied coefficient-wise.